Jaydeb Sarker

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Research Interests and Expertise

Human Aspects of Software Engineering (SE), Empirical Analysis, GenAI for Programming Education, Explainable AI for SE, Trustworthiness of AI for SE, Open Source Supply Chain, Code Review, Natural Language Processing

Education

Wayne State University — Detroit, MI
Ph.D. in Computer Science, GPA: 3.93/4.0 — *August 2019 – August 2024*
Dissertation Title: Identification and Mitigation of Toxic Communications Among Open Source Software Developers.

Advisor: Dr. Amiangshu Bosu

Wayne State University — Detroit, MI
Masters in Computer Science, GPA: 3.90/4.0 — *August 2019 – August 2022*

Rajshahi University of Engineering and Technology — Rajshahi, Bangladesh
BSc. in Computer Science and Engineering, CGPA: 3.71/4.0 (7^{th}/54) — *January 2012 – October 2016*

Employment

Assistant Professor (Tenure Track) — August 2024 – Present
Department of Computer Science, University of Nebraska at Omaha — *Omaha, NE*

Graduate Teaching Assistant — August 2019 – August 2022, May 2023 - May 2024
Wayne State University — *Detroit, MI*

Summer Camp Instructor — July 2024 – August 2024, June 2022- August 2022
Wayne State University — *Detroit, MI*

Thomas C Rumble Graduate Fellow — August 2022 – May 2023
Wayne State University — *Detroit, MI*

Lecturer — January 2017 – July 2019
University of Information Technology and Sciences — *Dhaka, Bangladesh*

Research Internship — September 2017- November 2017
Otto-Von-Guericke-Universität — *Magdeburg, Germany*

Research Grant

- Nebraska Research Initiative (NRI 2026)- Title: "From Code Review to Secure Software: Data Foundations for Vulnerability Detection", PI: Jaydeb Sarker, Co-PI: Harvey Siy, Kshitiz Aryal, Robert Dyer, Amount: 99.99 K USD, Duration: July 2026- June 2027.

- UCRCA Grant 2026- Title: "Investigating the Impact of Generative AI in Asynchronous Introductory Software Engineering Course: A Pilot Study", The University Committee on Research and Creative Activity (UCRCA), ORCA, UNO. Role: Sole-PI, Amount: 10K USD. Duration: June 2026- May 2027.

Honors and Awards

- UCRCA Award from ORCA, UNO, 2026
- Distinguished Reviewer Award from MSR Conference 2026 (Technical/Research Track)
- ACM SIGSOFT CAPS award for attending FSE-2026, FSE-2025
- Outstanding Graduate Teaching Assistant -2024, Computer Science Department at WSU
- NSF Travel Award for attending ESEM-2023
- ACM SIGSOFT CAPS award for attending FSE-2023, ESEM-2023
- NSF Travel Award for attending the Midwest PL Summit 2023
- WSU Graduate Travel Award 2023 (FSE-2023), 2022 (ASE-2022)
- Thomas C. Rumble University Graduate Fellowship for 2022-23 Academic Year (9-month fellowship with full tuition fee waiver), Wayne State University
- Deutscher Akademischer Austauschdienst (DAAD) scholarship, Germany- 2017
- Technical scholarship in RUET for the outstanding results from 2012 to 2016.

Peer Reviewed Research Publications

Journal Papers

J1. **[TOSEM-25]** Sayma Sultana, **Jaydeb Sarker**, Farzana Israt, Rajshakhar Paul, and Amiangshu Bosu. Automated identification of sexual orientation and gender identity discriminatory texts from issue comments. *Accepted in the ACM Transactions on Software Engineering and Methodology*, july 2025 **Impact Factor: 6.6**.
Accepted to Present in the Journal First Track of FSE 2026, Montreal, Canada.

J2. **[TOSEM-23]** **Jaydeb Sarker**, Asif Kamal Turzo, Ming Dong, and Amiangshu Bosu. Automated identification of toxic code reviews using toxicr. *ACM Transactions on Software Engineering and Methodology (TOSEM)*, 32(5), july 2023 **Impact Factor: 6.6**.
Accepted and Presented in the Journal First Track of ESEC/FSE-2023, San Francisco, USA.

Peer Reviewed Conference Papers

C1. **[FSE-26]** Md Awsaf Alam Anindya, Showvik Biswas, Anindya Iqbal, **Jaydeb Sarker**, and Amiangshu Bosu. Toxishield: Enhancing developer collaboration through real-time toxicity filtering. *Proceedings of the ACM on Software Engineering*, (FSE), 2026 **Acceptance Rate (211/920= 22.9%)**.

C2. **[FSE-25]** **Jaydeb Sarker**, Asif Kamal Turzo, and Amiangshu Bosu. The landscape of toxicity: An empirical investigation of toxicity on github. *Proceedings of the ACM on Software Engineering*, 2(FSE):623–646, 2025. **Direct Accepted, Acceptance Rate Without Major Rev: 11.5% (70/607), Overall (22%)**.

C3. **[ESEM-23]** **Jaydeb Sarker**, Sayma Sultana, Steven R Wilson, and Amiangshu Bosu. Toxispanse: An explainable toxicity detection in code review comments. In *2023 ACM/IEEE International Symposium on Empirical Software Engineering and Measurement (ESEM)*, pages 1–12. IEEE, 2023. **Acceptance Rate: 29%.**

C4. **[ESEM-23]** Asif Kamal Turzo, Fahim Faysal, Ovi Poddar, **Jaydeb Sarker**, Anindya Iqbal, and Amiangshu Bosu. Towards automated classification of code review feedback to support analytics. In *2023 ACM/IEEE International Symposium on Empirical Software Engineering and Measurement (ESEM)*, pages 1–12. IEEE, 2023. **Acceptance Rate: 29%.**

C5. **[APSEC-20]** **Jaydeb Sarker**, Asif Kamal Turzo, and Amiangshu Bosu. A benchmark study of the contemporary toxicity detectors on software engineering interactions. In *2020 27th Asia-Pacific Software Engineering Conference (APSEC)*, pages 218–227. IEEE, 2020. **Acceptance Rate: 37%.**

Workshop / Short Papers

SP1 **[Full Research Paper at LLMSC-26, Co-Located FSE-26]** Fariha Tanjim Shifat, Hariswar Baburaj, Ce Zhou, **Jaydeb Sarker**, and Mia Mohammad Imran. Llm-enabled open-source systems in the wild: An empirical study of vulnerabilities in github security advisories. In *Proceedings of the 34th ACM International Conference on the Foundations of Software Engineering*, FSE Companion '26. ACM New York, NY, USA, 2026

SP2 **[FSE-Poster-26]** Md Awsaf Alam Anindya, Showvik Biswas, Anindya Iqbal, **Jaydeb Sarker**, and Amiangshu Bosu. Real-time toxicity filtering for open-source code reviews. In *Proceedings of the 34th ACM International Conference on the Foundations of Software Engineering*, FSE Companion '26. ACM New York, NY, USA, 2026

SP3 **[NLBSE-26]** Satyanarayana Chowdary Kadiyala, **Jaydeb Sarker**, and Bianca Trinkenreich. How should self-deprecation comments be classified? a toxicity analysis study on zephyr. In *Proceedings of the Sixth ACM/IEEE International Workshop on NL-based Software Engineering (Co-Located with ICSE)*, 2026 (Accepted)

SP4 **[FSE-IVR-25]** Mia Mohammad Imran* and **Jaydeb Sarker***. "silent is not actually silent": An investigation of toxicity on bug report discussion. In *The ACM International Conference on the Foundations of Software Engineering (FSE), Ideas, Visions and Reflections Track*, 2025

SP5 **[ASE-22]** **Jaydeb Sarker**. 'who built this crap?' developing a software engineering domain specific toxicity detector. *Student Research Competition on the International Conference on Automated Software Engineering (ASE), Rochester, MI, USA*, pages 1–3, 2022.

SP6 **[ASE-22]** **Jaydeb Sarker**. Identification and mitigation of toxic communications among open source software developers. *Doctoral Symposium on the International Conference on Automated Software Engineering (ASE), Rochester, MI, USA*, pages 1–5, 2022.

SP7 [**ESEM-21**] Sayma Sultana, **Jaydeb Sarker**, and Amiangshu Bosu. A rubric to identify misogynistic and sexist texts from software developer communications. In *Proceedings of the 15th ACM/IEEE International Symposium on Empirical Software Engineering and Measurement (ESEM)*, pages 1–6, 2021.

Reviewing/Community Services

Grant Reviews

- NSF Graduate Research Fellowship Program- 2025, 2026

Journal Reviewer

1. IEEE Transactions on Software Engineering (TSE): 2024 Present
2. ACM Transactions on Software Engineering and Methodology (TOSEM): 2024- Present
3. Empirical Software Engineering (EMSE): 2025, 2026
4. IEEE Transactions on Reliability (T-REL): 2025
5. IEEE Software: 2025
6. Software Quality Journal: 2022-Present
7. e-Informatica Software Engineering Journal: 2024-Present

Conference Program Committee/Reviewer

1. PC Member at the 41st IEEE/ACM International Conference on Automated Software Engineering (ASE) - Research Track- 2026 (A* conference)
2. PC Member at the ACM International Conference on the Foundations of Software Engineering-Software Engineering Education (FSE-SEET) Track- 2026
3. PC Member at the International Conference on Software Maintenance and Evolution (ICSME)-Research Track- 2026
4. PC Member at the Mining Software Repositories Conference (MSR)- Research Track- 2026
5. PC Member at LLMTrust 2026 (Co-Located workshop with FSE 2026)
6. International Conference on Software Engineering (ICSE): Artifact Evaluation Track: 2025, 2024
7. Junior PC Member at the Mining Software Repositories Conference (MSR) 2025, 2024, 2023
8. International Conference on Program Comprehension (ICPC): Early Research Achievements (ERA Track) 2025
9. The Evaluation and Assessment in Software Engineering (EASE): Emerging Results Track 2025, Learnings and Reflections Track 2025
10. Software Analysis, Evolution and Reengineering (SANER) : Tool Demo Track 2024
11. MSR 2021: Shadow Program Committee member
12. Additional Reviewers within the ICSE 2021- Tool Demonstrations Track

Student Volunteers

1. 37th IEEE/ACM ASE 2022, In-person Conference at Oakland Center, Michigan.
2. 44th ICSE-2022, In-person Conference at Pittsburgh, PA, USA.
3. 36th IEEE/ACM ASE 2021, Virtual (Original: Melbourne, Australia).
4. 43rd ICSE-2021, Virtual Conference (Original: Madrid, Spain).

Teaching

University at Nebraska at Omaha, USA

- CSCI 4830: Introduction to Software Engineering, Spring 2025, 2026, Fall 2024
- CSCI 4500: Operating Systems, Spring 2026
- CSCI 8920: Empirical Software Engineering, Spring 2025 (Graduate Level)
- CSCI 3320 Data Structures, Fall 2025

Wayne State University, MI, USA

Course/Lab Instructor

- CSC 4110 - Software Engineering Lab: Winter 2021, Summer 2021, Winter 2022, Summer 2022, Fall 2023, Winter 2024
- CSC 4110 - Software Engineering Lecture: Summer 2022
- CSC 4420 - Computer Operating Systems (Theory+Lab) - Summer 2023
- CSC 1100 - Introduction to Problem Solving and Programming Lab: Fall 2019, Winter 2020, Summer 2020, Fall 2020
- CSC 1101- Introduction to Problem Solving and Programming Lecture: Summer 2020

Teaching Assistant

- BE 1500: Introduction to Programming and Computation for Engineers

University of Information Technology and Sciences, Dhaka, Bangladesh

Instructor

- C Programming, Data Structure, Algorithms, Software Engineering (January 2017-July 2019)

Student Advising

Graduate Students, UNO

- Steve Saunders (PhD in CIS): Fall 2025
- Rudra Sarkar (MS in CS): Fall 2025

K-12 Interns at College of IS&T UNO

- Cassandra Lucas: Summer 2025
- Anshul Bihani: Summer 2025

Student Awards

- UNO GRACA award (5000 USD for Summer 2026)- Steve Saunders
- UNO GRACA award (5000 USD for Summer 2026)- Rudra Sarkar

Developed Software/Tools for SE Research

ToxiShield | *Java Script, Python* [GitHub]
A real-time detoxification tool for GitHub

ToxiCR | *Python, Tensorflow* [GitHub]

- BERT-base model achieved an 89% F1-score and outperformed other SOTA toxicity detectors

ToxiSpanSE | *Python, Tensorflow, PyTorch* [GitHub]

- RoBERTa model achieved 88% F1 score for toxic class tokens

Research and Technical Skills

Programming: Python, Java, C, C++, SQL, MATLAB, Java Script
Statistical Data Analysis: Empirical Analysis of Software Engineering, Regression Modeling, Bootstrapping in Regression, Qualitative Analysis
NLP and ML: Classification, Deep Neural Models, Transformers, BERT, RoBERTa, XLNet, Token Level Text Classification, Huggingface Transformers, Explainability of Transformers

Outreach Programs and Extra Curricular Activities

- Lead Instructor of Summer Camp program at Wayne State University, 2022 and 2024 (worked with 4-12 grade students)
- Coach of Analytical Programming Contest Team-2018, UITS, Dhaka, Bangladesh

References

Available upon request